Shop kart Analysis Project

Objective: The objective of this project is to analyse Shop Kart retail sales data to identify sales trends, customer behaviour, profitable categories, and loss-making products. The analysis helps management make data-driven decisions to improve profitability and optimize operations.

1.Dataset Information:

1.Total Records: 60,000 Records

2.Columns:
Order_ID,Order_Date,Delivery_Date,Customer_ID,Gender,Age,City,State,Product_ID,Product_Name,Category,Cost_Price,Selling_Price,Discount_%,Quantity,Revenue,Profit,Payment_Method,Order_Status.

Date_Table:Date,Year,Month_Number,Month

2.Tool Used:

1.Excel → Data understanding

2.Python → Data cleaning

3.SQL → Data analysis

4. Power BI → Dashboard visualization

3. Data Cleaning Steps:

1.Removed duplicate records

2.Checked missing values

3.Converted date format

4.Verified negative profit values (kept for loss analysis)

5. Converted dataset into a structured format for SQL analysis.

4.SQL Data Analysis:

1. Which category is making high profit contribution.

```
SELECT Category,

SUM(Profit) AS Total_Profit,

SUM(Profit) * 100.0 /

(SELECT SUM(Profit) FROM shopkart_data) AS Profit_Contribution_Percent
```

```
FROM shopkart_data
GROUP BY Category
ORDER BY Total_Profit DESC;
```

Results | Messages

	Category	Total_Profit	Profit_Contribution_Percent
1	Home	15056437.672503	41.8353537950272
2	Electronics	14903020.637229	41.4090739479296
3	Clothing	6030287.93759783	16.7555722570432

Categories with the highest profit contribution are critical for business growth and strategic investments.

2. Which category generates the most sales.

```
SELECT Category,
SUM(Revenue) AS Total_Sales
FROM shopkart_data
GROUP BY Category
ORDER BY Total_Sales DESC;
```

Results | Messages

	Category	Total_Sales
1	Home	111301288.409317
2	Electronics	110325314.502655
3	Clothing	102210065.975677

This helps identify core product categories contributing to revenue growth.

3. Which payment method made high profit and discount.

```
select Payment_Method,
sum(Profit) as Payment_method_wise_profit,
sum(Discount) as Discount
from shopkart_data
group by Payment_Method
order by sum(Profit)  ASC;
```

	Payment_Method	Payment_method_wise_profit	Discount
1	Debit Card	8954931.71864674	1495.30002823099
2	COD	8957661.45923516	1509.20002879947
3	Credit Card	9029671.12039191	1504.20002867654
4	UPI	9047481.94905599	1496.45002854243

Understanding payment preferences helps improve customer checkout experience and payment partnerships.

4. What is the overall business performance?

```
SELECT

SUM(Revenue) AS Total_Revenue,

SUM(Profit) AS Total_Profit,

(SUM(Profit) * 100.0 / SUM(Revenue)) AS Profit_Margin

FROM shopkart_data;
```

Results | Messages

	Total_Revenue	Total_Profit	Profit_Margin
1	323836668.88765	35989746.2473298	11.1135488056221

This KPI provides a high-level overview of company performance. Management uses this to evaluate overall profitability and financial health.

5. How many orders are generating losses?

```
SELECT

COUNT(*) AS Negative_Orders,

SUM(Profit) AS Total_Loss

FROM shopkart_data

WHERE Profit < 0;
```

	Negative_Orders	Total_Loss
1	5897	-1221679.93004076

This helps identify the scale of loss-making transactions, allowing management to investigate pricing, discount policies, or operational inefficiencies.

6. Which product categories generate the most losses?

```
SELECT Category,
```

```
COUNT(*) AS Loss_Orders,
SUM(Profit) AS Total_Loss
FROM shopkart_data
WHERE Profit < 0
GROUP BY Category
ORDER BY Total_Loss ASC;
```

	Category	Loss_Orders	Total_Loss
1	Clothing	5193	-1204794.3100486
2	Home	358	-8888.01000084169
3	Electronics	346	-7997.60999131575

Categories with frequent losses may require price adjustments or discount optimization.

7.Are Discount Causing losses?

```
SELECT Category,
AVG(Discount) AS Avg_Discount,
SUM(Profit) AS Total_Loss
FROM shopkart_data
WHERE Profit < 0
GROUP BY Category
ORDER BY Avg_Discount DESC;
```

Results | Messages

	Category	Avg_Discount	Total_Loss
1	Clothing	0.178962068182814	-1204794.3100486
2	Electronics	0.100000001490116	-7997.60999131575
3	Home	0.100000001490116	-8888.01000084169

High discounts often reduce profitability. Businesses can optimize discount strategies based on this analysis.

8.Which Product Generate Highest Loss?

```
SELECT TOP 10
Product_Name,
SUM(Profit) AS Total_Loss
```

```
FROM shopkart_data
WHERE Profit < 0
GROUP BY Product_Name
ORDER BY Total_Loss ASC;
```

	Product_Name	Total_Loss
1	T-Shirt	-305121.859752849
2	Jeans	-303888.980836064
3	Jacket	-302802.489370029
4	Shirt	-292980.980089657
5	Chair	-2360.81000358798
6	Smartwatch	-2311.10000689328
7	Headphones	-2238.50997920334
8	Mixer	-2219.44999101758
9	Table	-2168.6499979049
10	Sofa	-2139.10000833124

Products generating frequent losses should be reviewed for pricing or supplier cost adjustments.

9.Which Customer Contribute the most Profit?

```
SELECT TOP 10
Customer_ID,
SUM(Profit) AS Total_Profit,
COUNT(Order_ID) AS Total_Orders
FROM shopkart_data
GROUP BY Customer_ID
ORDER BY Total_Profit DESC;
```

	Customer_ID	Total_Profit	Total_Orders
1	CUST7569	17714.640045166	10
2	CUST2156	15141.0999603271	11
3	CUST7340	13063.8899993896	10
4	CUST10585	12921.4299278259	11
5	CUST4767	12817.8399658203	7
6	CUST8400	12567.0800476074	9
7	CUST8553	12458.9500732422	7
8	CUST69	12341.3699035645	9
9	CUST13008	12087.15990448	8
10	CUST11867	11950.1100845337	10

High-value customers should be targeted with loyalty programs and personalized marketing campaigns.

10. What percentage of profit is lost due to negative transactions?

```
SELECT
(SUM(CASE WHEN Profit < 0 THEN Profit ELSE 0 END) * 100.0
/ SUM(Profit)) AS Loss_Percentage
FROM shopkart_data;
```

	Loss_Percentage
1	-3.39452221097946

This metric helps quantify the overall financial impact of loss-making orders.

11. What percentage of orders are generating losses?

```
SELECT
COUNT(CASE WHEN Profit < 0 THEN 1 END) * 100.0
/ COUNT(*) AS Loss_Order_Percentage
FROM shopkart_data;
```

	Loss_Order_Percentage
1	9.828333333333

A high loss-order percentage indicates inefficient pricing or operational problems.

12.What are the Monthly Profit trends?

```
SELECT
YEAR(Order_Date) AS Year,
MONTH(Order_Date) AS Month,
SUM(Profit) AS Monthly_Profit
FROM shopkart_data
GROUP BY YEAR(Order_Date), MONTH(Order_Date)
ORDER BY Year, Month;
```

Results | Messages

	Year	Month	Monthly_Profit
1	2024	1	2984200.37087618
2	2024	2	2786294.4122228
3	2024	3	3001422.02862144
4	2024	4	3065105.91941936
5	2024	5	3032058.59563993
6	2024	6	2871580.99775743
7	2024	7	3055863.99857119
8	2024	8	3092069.28930805
9	2024	9	2962551.40068391
10	2024	10	3116159.41185708
11	2024	11	2872950.59231981
12	2024	12	3149489.23005261

Monthly analysis helps identify seasonal demand patterns and allows management to adjust marketing and inventory strategies.

13.Which city generate highest revenue?

```
SELECT City,
SUM(Revenue) AS Total_Revenue
FROM shopkart_data
GROUP BY City
ORDER BY Total_Revenue DESC;
```

	City	Total_Revenue
1	Mumbai	65828808.2969818
2	Bangalore	65432488.1027069
3	Hyderabad	64380906.9358063
4	Delhi	64174180.4989014
5	Pune	64020285.0532532

High-performing cities represent strong markets for business expansion and targeted promotions.

14.Which generate the highest sales?

```
SELECT TOP 10 Product_Name,
SUM(Revenue) AS Total_Revenue
FROM shopkart_data
GROUP BY Product_Name
ORDER BY Total_Revenue DESC;
```

	Product_Name	Total_Revenue
1	Mixer	28288901.1173859
2	Laptop	28144570.2325897
3	Chair	27813430.0092468
4	Headphones	27801955.2151184
5	Table	27737134.7638245
6	Sofa	27461822.5188599
7	Mobile	27194328.5521698
8	Smartwatch	27184460.5027771
9	Jacket	25839408.77005
10	T-Shirt	25636484.8728485

These products are key revenue drivers and should be prioritized for inventory management and marketing campaigns.

15.Top 3 Products each category generate high revenue?

```
WITH ranked_products AS (
SELECT
Category,
Product_Name,
SUM(Revenue) AS Total_Revenue,
RANK() OVER(PARTITION BY Category ORDER BY SUM(Revenue) DESC) AS rnk
FROM shopkart_data
GROUP BY Category, Product_Name
)
SELECT *
FROM ranked_products
WHERE rnk <= 3;
```

	Category	Product_Name	Total_Revenue	rnk
1	Clothing	Jacket	25839408.77005	1
2	Clothing	T-Shirt	25636484.8728485	2
3	Clothing	Shirt	25620954.0669708	3
4	Electronics	Laptop	28144570.2325897	1
5	Electronics	Headphones	27801955.2151184	2
6	Electronics	Mobile	27194328.5521698	3
7	Home	Mixer	28288901.1173859	1
8	Home	Chair	27813430.0092468	2
9	Home	Table	27737134.7638245	3

4.Power Bi Dashboard:

1.Excutive Summary Dashboards

Insights:

1. **Monthly revenue fluctuates across the year, indicating seasonal sales patterns. This allows management to adjust marketing campaigns and inventory planning.**
2. **Electronics and Home categories contribute the majority of profits, while Clothing contributes a smaller share.**
3. **Maharashtra generates the highest profit among all states, suggesting a strong market presence in that region.**
4. **Nearly 5.8K orders generate losses, indicating potential pricing or discount issues**

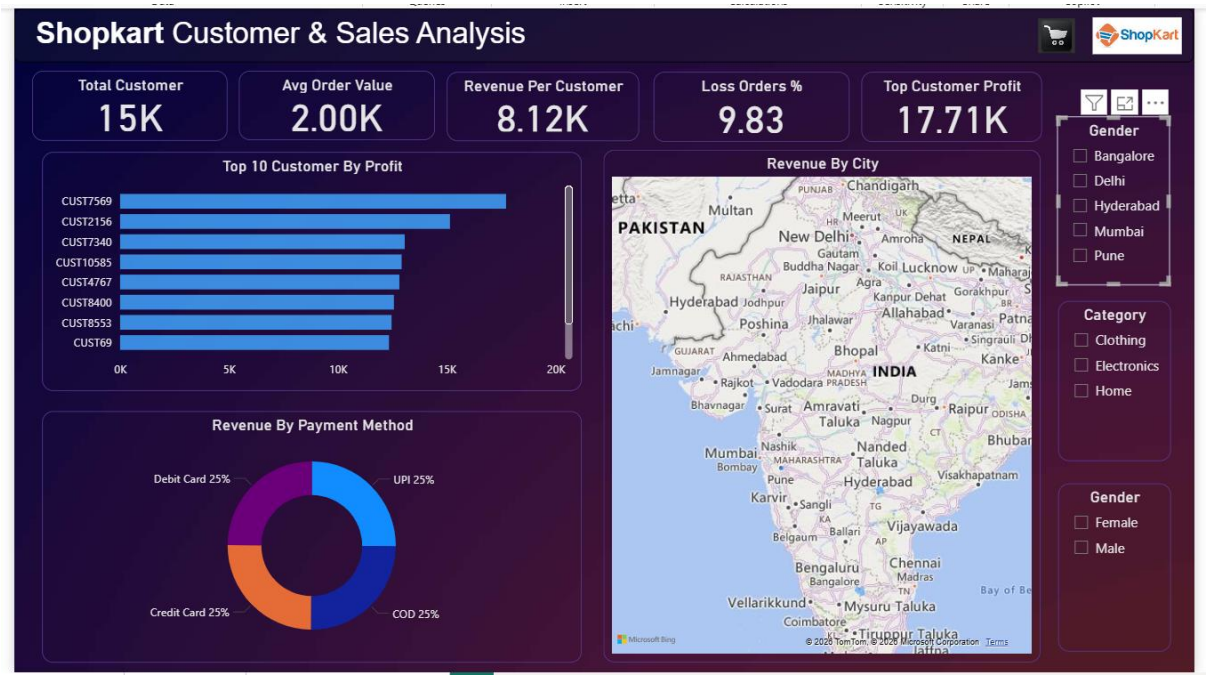
2.Loss Analysis

1. Most losses come from the Clothing category, which may indicate aggressive discounting or high supplier costs.

2. Products like Jackets, T-shirts, and Jeans generate the highest losses. These products require pricing or supplier negotiation review.

3. The discount vs profit scatter plot shows that higher discounts are strongly associated with negative profit. This insight helps management optimize discount policies.

3.Customer & Sales Analysis



1. **Average order value is around 2K, which helps the business estimate expected customer spending.**
2. **A small group of customers contributes a significant share of profit. These customers should be targeted with loyalty programs and personalized promotions.**
3. **Major cities like Mumbai, Delhi, Bangalore, and Hyderabad contribute the highest revenue. These regions should be prioritized for marketing campaigns and inventory distribution.**
4. **UPI and card payments contribute the majority of revenue, indicating a strong digital payment preference among customers.**

5.Key Business Insights:

1.Seasonal Sales Trend Insight

Insight:
The monthly revenue trend shows fluctuations throughout the year, indicating seasonal demand patterns.

Business Impact:
Understanding seasonal trends allows the company to:

- Increase inventory before high-demand months
- Launch promotional campaigns during peak sales periods

- Optimize supply chain planning

Recommendation:
Plan marketing campaigns and stock management based on seasonal demand patterns.

2.High Loss Orders Impact

Insight:
Approximately 5,897 orders are generating losses, which represents a significant portion of total transactions.

Business Impact:
Loss-generating orders directly reduce overall profitability and may indicate:

- Excessive discounting
- High shipping or supplier costs
- Pricing inefficiencies

Recommendation:
Review pricing strategies and discount policies to reduce loss-making transactions.

3. Category Performance Insight

Insight:
The Electronics and Home categories contribute the highest revenue and profit, while the Clothing category generates lower profitability and higher losses.

Business Impact:
This indicates that some product categories perform significantly better than others.

Recommendation:

- Increase marketing investment in high-performing categories
- Reevaluate pricing and cost structure in the Clothing category

4. High-Value Customer Contribution

Insight:
A small group of customers contributes a significant share of total profit.

Business Impact:
These high-value customers are critical for revenue growth and long-term business sustainability.

Recommendation:
Implement customer loyalty programs, personalized promotions, and retention strategies to maintain these valuable customers.

5. Regional Market Performance

Insight:
States such as Maharashtra and Karnataka generate the highest profits, indicating strong regional demand.

Business Impact:
These regions represent key markets where the company has strong customer engagement and sales performance.

Recommendation:

- Expanding marketing efforts in high-performing regions
- Strengthening distribution networks in profitable states.

6.Conclusion: The Shop Kart Sales Analytics project analysed retail sales data to understand business performance, customer behaviour, and product profitability. The analysis identified key revenue-driving categories, loss-generating products, and high-value customers. The Power BI dashboard provides interactive insights into sales trends, regional performance, and discount impact on profit. These insights help the business make data-driven decisions to improve profitability, optimize pricing strategies, and strengthen customer engagement.